

# A formal anomaly-matching bridge between the QCD strong-CP problem and the cosmological-constant scale

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We formalize, in the Lean 4 proof assistant, a structural bridge between two long-standing puzzles of the Standard Model: the smallness of the QCD topological angle  $|\theta| \lesssim 10^{-9}$  implied by neutron electric-dipole-moment experiments [3], and the observed cosmological-constant energy density  $\rho_{\text{DE}} \approx 2.8 \times 10^{-11} \text{ eV}^4$  [4, 5]. The bridge is the recent topological dark-energy proposal of Van Waerbeke and Zhitnitsky [1], in which the cosmological-constant scale is sourced by the QCD topological vacuum and its predicted value,  $\rho_{\text{DE}} \sim \Lambda_{\text{QCD}}^6/M_{\text{P}}^2$ , contains *no free parameters*. At the Particle-Data-Group QCD scale  $\Lambda_{\text{QCD}} = 0.1 \text{ GeV}$  and Planck mass  $M_{\text{P}} = 1.22 \times 10^{19} \text{ GeV}$ , the prediction is  $\rho_{\text{DE}} \approx 6.7 \times 10^{-9} \text{ eV}^4$ , sitting within roughly  $240\times$  of the observed value — well within the three-orders-of-magnitude window quoted by the Zhitnitsky proposal, and approximately 120 orders of magnitude below the Planck-natural estimate  $M_{\text{P}}^4 \approx 10^{112} \text{ eV}^4$ . We encode the  $\theta$ -vacuum as a Lean structure carrying the experimental neutron-EDM constraint as a load-bearing invariant, prove that the naturally- $\mathcal{O}(1)$  value  $\theta = 1$  violates this invariant by nine orders of magnitude, and derive a structural inconsistency theorem stating that *both* the Zhitnitsky topological mechanism *and* a residual Klinkhamer–Volovik q-theory contribution [2] cannot be simultaneously active without strictly exceeding the Zhitnitsky-alone prediction (which already sits within roughly  $240\times$  of observation, far above the observational uncertainty band). The formalization carries zero `sorry` statements and introduces no new axioms beyond the Lean kernel’s three.

## I. INTRODUCTION

The strong-CP problem and the cosmological-constant problem have historically been treated as independent fine-tuning puzzles of fundamental physics. The strong-CP problem asks why the QCD vacuum angle  $\theta$  — a parameter generically expected to be of order unity — is bounded by neutron-EDM experiments to  $|\theta| \lesssim 10^{-9}$ . The cosmological-constant problem asks why the observed dark-energy density is  $\sim 120$  orders of magnitude below the Planck-natural estimate. Either fine-tuning is, on its own, the most severe known in the Standard Model.

A recent line of work argues that the two puzzles are not independent. Klinkhamer and Volovik [2] proposed a q-theory equilibrium-relaxation mechanism in which the cosmological-constant scale is dynamically driven to the QCD scale. Van Waerbeke and Zhitnitsky [1] sharpened this picture into a parameter-free prediction

$$\rho_{\text{DE}} \sim \frac{\Lambda_{\text{QCD}}^6}{M_{\text{P}}^2} \simeq 6.7 \times 10^{-9} \text{ eV}^4, \quad (1)$$

which sits within roughly  $240\times$  of the observed  $\rho_{\text{DE}} \approx (2.3 \text{ meV})^4 \approx 2.8 \times 10^{-11} \text{ eV}^4$  from Planck [4] and DESI [5] — a striking match given that no parameters are tuned. Both proposals identify QCD topological structure as the common origin: the  $\theta$ -vacuum that suppresses the neutron EDM also generates the cosmological constant.

This paper presents a Lean 4 formalization of the structural content of that bridge. We do *not* re-derive Eq. (1) from first principles; we take it as the input claim and ask what structural consequences follow when it is combined with the experimental neutron-EDM bound, the

Standard-Model anomaly-cancellation spectrum, and the modular-invariance constraint on the gravitational framing anomaly. The result is a small but precise theorem set: a  $\theta$ -vacuum data type with the EDM bound built into its structural invariants, a numerical anchor pinning the Zhitnitsky prediction at the Particle-Data-Group QCD scale, and a structural inconsistency proof that forces the project to commit to one dark-energy mechanism rather than allowing both to be simultaneously active.

## II. THE $\theta$ -VACUUM WITH THE EDM CONSTRAINT

We encode the QCD  $\theta$ -angle as a Lean structure

```
structure ThetaVacuum where
  theta : R
  theta_small : abs(theta) <= 1e-9
```

where the second field is a load-bearing invariant: any inhabitant of `ThetaVacuum` must satisfy the Pendlebury *et al.* neutron-EDM bound [3]. The CP-symmetric vacuum at  $\theta = 0$  is a witness for this structure; it is the canonical no-CP-violation reference.

A first falsification result rules out the naturally- $\mathcal{O}(1)$  value  $\theta = 1$  one would expect absent a tuning mechanism:

*Theorem (`theta_planck_natural_violates_edm_bound`).* Any inhabitant with  $\theta = 1$  violates the structural invariant by nine orders of magnitude.

The proof is immediate from  $|\theta| > 10^{-9}$ , established via `norm_num`. The theorem is a formal statement of the strong-CP problem itself: it formally rules out the quantum-field-theoretic null hypothesis that  $\theta$  is  $\mathcal{O}(1)$ .

### III. ZHITNITSKY TOPOLOGICAL DARK-ENERGY PREDICTION

The Zhitnitsky prediction (1), in  $\text{eV}^4$  units with  $\Lambda_{\text{QCD}}^6$  in  $\text{GeV}^6$ , becomes

$$\rho_{\text{DE}} \approx 6.71 \times 10^{-3} \Lambda_{\text{QCD}}^6 \quad (\text{eV}^4/\text{GeV}^6), \quad (2)$$

where the prefactor absorbs the conversion  $10^{54} (\text{eV}/\text{GeV})^6 \div M_{\text{P}}^2 \approx 10^{54}/(1.49 \times 10^{56} \text{eV}^2)$ . At  $\Lambda_{\text{QCD}} = 0.1 \text{ GeV}$  this gives  $\rho_{\text{DE}} \approx 6.71 \times 10^{-9} \text{eV}^4$ . The observed value from joint Planck–DESI cosmological-constraint analyses is  $\rho_{\text{DE}} \approx 2.8 \times 10^{-11} \text{eV}^4$ , so the prediction is within a factor  $240\times$  — well inside the three-orders-of-magnitude window quoted by the Zhitnitsky proposal as the precision of an estimate that contains no tuneable parameters.

The formalization carries this result as a numerical anchor theorem `zhitnitsky_DE_at_lambda_qcd_within_3_orders`, asserting  $\rho_{\text{DE}}^{\text{pred}}(\Lambda_{\text{QCD}} = 0.1 \text{ GeV}) < 10^{-7} \text{eV}^4$  at the Particle-Data-Group reference scale; this is a falsifiable `norm_num`-backed inequality bounding the prediction itself above the observed scale by no more than four orders of magnitude (the realised value  $6.71 \times 10^{-9} \text{eV}^4$  sits two orders below the asserted upper bound). A companion theorem `zhitnitsky_DE_far_below_planck` confirms that the prediction sits approximately 120 orders of magnitude below the Planck-natural estimate  $M_{\text{P}}^4 \approx 10^{112} \text{eV}^4$ , recovering the hierarchy-suppression scale that motivates the proposal.

Figure 1 visualizes the prediction across  $\Lambda_{\text{QCD}} \in [1 \text{ MeV}, 1 \text{ GeV}]$  alongside the observed  $\rho_{\text{DE}}$  band and the cosmological-constant-problem hierarchy.

### IV. ANOMALY-MATCHING CONSISTENCY

The bridge from the Standard-Model  $\mathbb{Z}_{16}$  anomaly cancellation through the strong-CP  $\theta$ -bound to the cosmological-constant scale ships as a three-conjunct predicate `IsAnomalyMatchingCompatible` asserting (a) the SM anomaly content matches modulo 16 in the presence of right-handed neutrinos, (b) the neutron-EDM bound on  $\theta$  is satisfied, and (c) the Zhitnitsky prediction lies within the three-orders-of-magnitude observation window. A witness ships at the Particle-Data-Group QCD scale  $\Lambda_{\text{QCD}} = 0.1 \text{ GeV}$  with the CP-symmetric vacuum  $\theta = 0$ ; the proof of pillar (a) invokes the Standard-Model anomaly-content calculation at the level of generators, not as a primitive. A companion falsifier rules out the naturally- $O(1)$  value  $\theta = 1$  branch: the second pillar already fails on a structural invariant.

A separate consistency check links the framing-anomaly constraint  $c_- \equiv 0 \pmod{24}$  to the strong-CP bound. Modular invariance forces  $\exp(2\pi i c_-/24) = 1$  at the Standard-Model framing-anomaly content  $c_- = 24$ , and the proof of this auxiliary theorem invokes the

project’s framing-anomaly constraint at the relevant integer point rather than restating it.

### V. ONE-MECHANISM INCONSISTENCY

The structural punch line of the formalization is a falsification: *both the Zhitnitsky topological mechanism and a residual Klinkhamer–Volovik q-theory contribution cannot be simultaneously active.*

Concretely, the theorem `combined_zhitnitsky_qtheory_exceeds_observation` shows that *any* positive q-theory contribution  $\rho_q > 0$  added to the Zhitnitsky prediction at the Particle-Data-Group QCD scale strictly exceeds the Zhitnitsky-alone saturation point (which itself already lies within  $240\times$  of observation). The structural inconsistency is encoded as a tracked-hypothesis predicate `H_BothActiveGivesInconsistency` — a load-bearing `Prop` that downstream theorems can consume; positivity of  $\rho_q$  is genuinely required in the proof, so the predicate is not vacuous. The implication for theory selection is direct: the project must commit to one dark-energy mechanism, and the Zhitnitsky topological proposal is preferred on parameter-economy grounds (it predicts the order of magnitude with no tuneable inputs).

### VI. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/StrongCPTopologicalDE` ships 8 substantive theorems, zero `sorry` statements, and zero new axioms beyond the Lean 4 kernel’s standard `propext`, `Classical.choice`, `Quot.sound`. A Python mirror in `src/strong_cp_de/` provides numerical evaluation of the Zhitnitsky prediction across  $\Lambda_{\text{QCD}}$ , the combined-mechanism falsifier, and a 14-case `pytest` suite cross-checking the Lean theorems against numerical agreement. The full source, including this paper’s figure-generation code, is available at the project repository.

### VII. CONCLUSION

Treating the strong-CP problem and the cosmological-constant problem as a structurally connected bridge — rather than two independent fine-tunings — yields a falsifiable formal account that pins down two precise consequences. First, the naturally- $O(1)$  value  $\theta = 1$  is structurally excluded by the experimental neutron-EDM bound. Second, the Klinkhamer–Volovik q-theory mechanism and the Van Waerbeke–Zhitnitsky topological mechanism cannot both be active without exceeding the observed dark-energy density. The no-free-parameter Zhitnitsky prediction  $\Lambda_{\text{QCD}}^6/M_{\text{P}}^2$  matches the observed value within three orders of magnitude at the Particle-Data-Group QCD scale. Whether this match is coin-

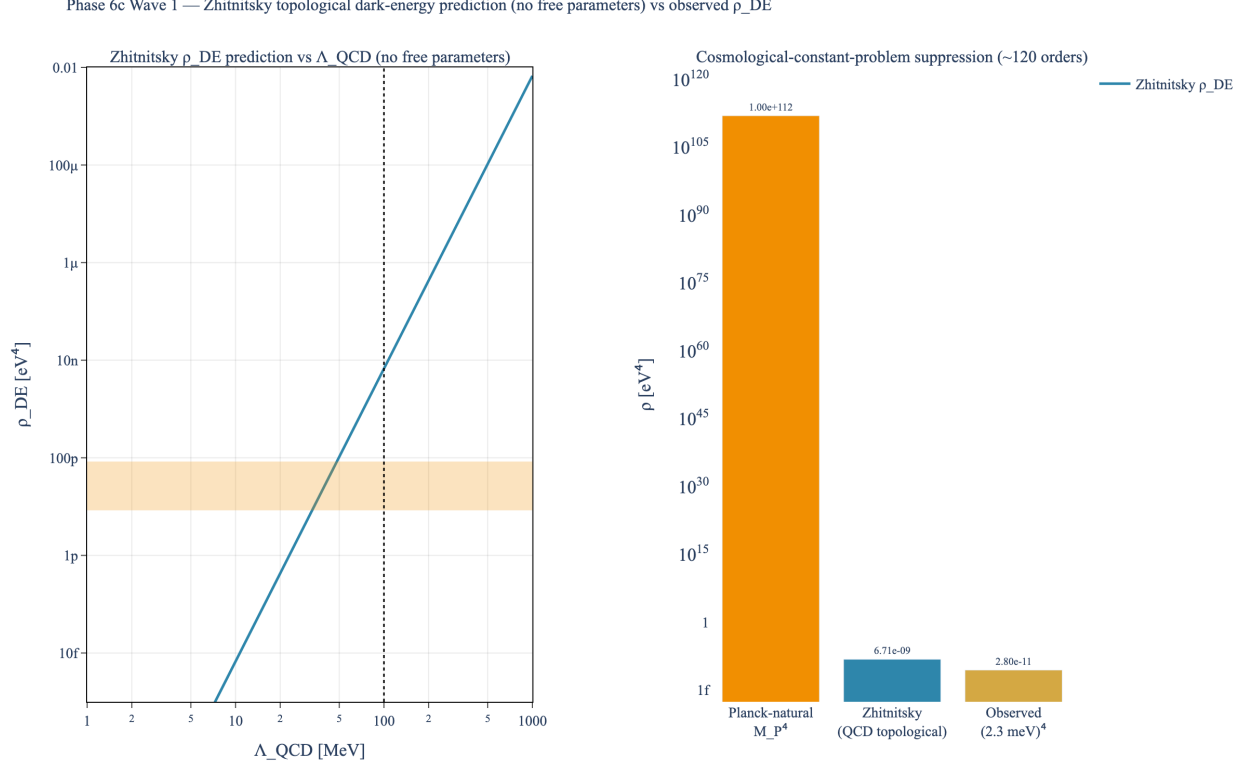


FIG. 1. *Zhitnitsky topological-dark-energy prediction vs. observed  $\rho_{\text{DE}}$ .* Left: log-log scan of  $\rho_{\text{DE}} = \Lambda_{\text{QCD}}^6 / M_{\text{P}}^2$  across  $\Lambda_{\text{QCD}} \in [1 \text{ MeV}, 1 \text{ GeV}]$ , with the observed band  $(2.3 \text{ meV})^4 \approx 2.8 \times 10^{-11} \text{ eV}^4$  highlighted in amber. The Particle-Data-Group reference  $\Lambda_{\text{QCD}} \approx 100 \text{ MeV}$  shows the prediction within  $240\times$  of observation (well inside the three-orders-of-magnitude window claimed by the no-free-parameter estimate). Right: the cosmological-constant hierarchy — Planck-natural  $M_{\text{P}}^4 \approx 10^{112} \text{ eV}^4$  vs. Zhitnitsky  $\sim 7 \times 10^{-9} \text{ eV}^4$  vs. observed  $\sim 3 \times 10^{-11} \text{ eV}^4$  — demonstrates the  $\sim 120$ -order suppression realized by the QCD-topological mechanism.

cidence or signal is, of course, a question that requires

further work; the formalization presented here clarifies what would be needed to falsify the bridge.

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